

REVENUE STRATEGY ANALYSIS | COMMERCIAL ANALYTICS

Which service investments *actually* move satisfaction — and for whom?

Identifying high-leverage satisfaction drivers by passenger segment to prioritize revenue-justified service investment across the commercial organization.

DATASET

103,904 Passenger Records

MODEL ACCURACY

91.2% (Decision Tree + Neural Network)

TOOLS

R · rpart · neuralnet · caret · ggplot2

ANALYST

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91.2%

Prediction Accuracy

88.99% sensitivity · 94.12% specificity

23%

Higher satisfaction likelihood from online boarding improvement

Strongest single revenue lever identified

2.1×

Business class wifi weight vs. economy

Premium service ROI signal

31%

Satisfaction gap: loyal vs. first-time flyers

Defines retention vs. acquisition spend logic

CORE ANALYSIS

Top Satisfaction Drivers — Ranked by Commercial Impact

1 TOP DRIVER

Online Boarding Experience

Strongest single predictor of satisfaction across all passenger segments. Passengers with high online

boarding scores show 23% higher likelihood of overall satisfaction.

Revenue implication: Digital boarding improvements deliver broad satisfaction lift at lower marginal cost than in-cabin service upgrades — the highest ROI intervention available.

2 SEGMENT DRIVER

Inflight Wifi Service

Business class passengers weight wifi satisfaction 2.1x higher than economy. Effect is minimal below a service quality threshold — passengers rating wifi 1–2 show near-zero satisfaction contribution.

Revenue implication: Wifi reliability is a premium product, not a parity feature. Framing wifi improvements as a Business class loyalty benefit (rather than a surcharge) could improve satisfaction and repeat booking simultaneously.

3 SEGMENT DRIVER

Travel Purpose (Personal vs. Business)

Travel purpose predicts satisfaction independently of all service ratings — personal travelers show higher sensitivity to every touchpoint, explaining their disproportionately lower baseline satisfaction.

Revenue implication: Segmentation-driven outreach strategy. Personal travelers are not lower-value passengers — they are higher-sensitivity customers requiring different service messaging and targeted experience improvements.

4 RETENTION DRIVER

Customer Loyalty Status

Loyal customers show 31% higher baseline satisfaction than first-time flyers, even when controlling for service ratings. The loyalty effect is structural, not service-driven.

Revenue implication: Satisfaction spend on loyal customers protects existing revenue more than it generates new revenue. First-flight experience improvements have higher long-run yield by shifting passengers into the loyal cohort.

SEGMENT ANALYSIS

Revenue Profile by Passenger Segment

 Business Travelers Highest revenue per seat · highest wifi sensitivity PRIORITY DRIVER Inflight wifi reliability SATISFACTION LEVEL ABOVE AVERAGE REVENUE ACTION Position premium wifi as loyalty perk, not surcharge	 Personal Travelers Highest churn risk · highest touchpoint sensitivity PRIORITY DRIVER Online boarding, then food & drink SATISFACTION LEVEL BELOW AVERAGE REVENUE ACTION Digital experience investment; segmented outreach strategy
 Loyal Customers 31% higher baseline · compounding retention value PRIORITY DRIVER Loyalty status independent of service ratings SATISFACTION LEVEL HIGHEST COHORT REVENUE ACTION Frame service investment as retention spend, not acquisition	 First-Time Flyers Highest conversion opportunity · lowest baseline SATISFACTION GAP –31% vs. loyal customers across all dimensions SATISFACTION LEVEL HIGHEST UPSIDE REVENUE ACTION First-flight experience improvements → loyalty cohort conversion

FEATURE IMPORTANCE

Relative Satisfaction Driver Strength (All Segments)

Online Boarding	96%
Inflight Wifi (Bus. Class)	84%
Type of Travel	79%
Cabin Class	73%
Inflight Entertainment	66%
Food & Drink	61%
Seat Comfort	58%
Customer Loyalty Status	54%

Relative importance derived from Decision Tree cost-complexity pruning. Bars represent normalized feature contribution; not raw coefficient values.

MODEL VALIDATION

Performance on Held-Out Test Data

METRIC	DECISION TREE	BENCHMARK
Accuracy	91.2%	≥85% target
Sensitivity (Recall)	88.99%	—
Specificity	94.12%	—
Kappa Score	0.8237	Strong agreement
Balanced Accuracy	91.55%	—

Why Decision Tree over Neural Network?

Both models reached comparable accuracy. The Decision Tree was selected as primary because satisfaction drivers can be traced directly to specific service features and explained to non-technical stakeholders — a requirement for any commercial analytics deliverable.

The Neural Network serves as a validation check: if both architectures agree on which passengers are dissatisfied, the finding is model-agnostic and more defensible in leadership presentations.

COMMERCIAL RECOMMENDATIONS

Three Revenue-Maximizing Actions

01

Prioritize digital boarding as the primary satisfaction investment

REVENUE IMPACT

High

Online boarding is the single highest-leverage intervention available — it affects all segments, delivers the largest satisfaction lift (23%), and requires no in-cabin capital expenditure. Recommend prioritizing digital boarding improvements in any service investment roadmap before in-cabin upgrades.

02

Reposition inflight wifi as a Business class loyalty benefit

Business class passengers weigh wifi 2.1x higher than economy. Rather than treating wifi as a commoditized add-on, integrating reliable wifi into Business class loyalty program value proposition — and investing in reliability improvements specifically for that cabin — maximizes return on the highest-yield segment.

REVENUE IMPACT

High

03

Target first-flight experience improvements to accelerate loyalty conversion

The 31% satisfaction gap between loyal and first-time flyers represents the highest-ROI long-run investment. First-time flyers who become loyal customers deliver compounding revenue across their lifetime. A targeted first-flight onboarding program — focused on the digital touchpoints this analysis identifies as highest-impact — is the clearest path from acquisition to retention revenue.

REVENUE IMPACT

Medium-High